

APPLICATION FOR ETHICAL CLEARANCE TO CONDUCT RESEARCH IN THE FACULTY OF ENGINEERING, BUILT ENVIRONMENT AND INFORMATION TECHNOLOGY

The Central University of Technology (CUT) Research Ethics and Integrity Policy applies to all undergraduate and post graduate students, and staff members who conduct research on CUT campuses and outside the campus. CUT policy bounds any person who wishes to conduct research with CUT students and/or staff but is not CUT affiliated to abide by the ethics framework. All CUT members who conduct research take responsibility to implement this Policy.

1. APPLICANT INFORMATION

1.1.	Title (Prof Dr /Mr /Mrs /Ms)	Mr							
1.2	Name(s) and Surname	Sizwe Marole							
1.3	Student / Staff number	222008531							
1.4	Department	Information Technology							
1.5	Campus	Bloemfontein campus							
1.6	Postal address								
1.7	Contact details	Office							
		Cell	0676759901						
		e-mail	222008531@stud.cut.ac.za						
1.8	Supervisor (s)/Project Leader								
1.9	Qualification registered for/Level of research	Please tick relevant option: <table border="1" style="width: 100%;"> <tr> <td>Masters qualification</td> <td>☐</td> </tr> <tr> <td>Doctorate</td> <td>☐</td> </tr> <tr> <td>Independent research (Non-qualification purposes)</td> <td>☐</td> </tr> </table>		Masters qualification	☐	Doctorate	☐	Independent research (Non-qualification purposes)	☐
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Doctorate	☐								
Independent research (Non-qualification purposes)	☐								
1.10	FRIC Approval Number (LS262a) (where applicable)								
1.11	Conflict of interest (Please underline/highlight): 1) Personal relationship Yes/ <u>No</u> 2) Financial benefit Yes/ <u>No</u>								

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	<i>If yes, please provide details:</i>

2. DETAILS OF THE STUDY

2.1	Approved/Proposed title of the study/project /dissertation/thesis
	Smart Streetlights for Smart Connectivity: A Solution to Bloemfontein's Urban Digital Divide
2.2	Research question(s)
	1. What are the benefits of using streetlights as Wi-Fi access points? 2. How can Wi-Fi-enabled streetlights contribute to the development of smart cities and greater digital inclusion in Bloemfontein? 3. What are the scalability prospects of streetlight-based Wi-Fi networks in large urban areas? 4. How does the lack of reliable internet access impact social and economic opportunities in Bloemfontein?
2.3	Aim and objectives of the study
	1. To assess the extent of the digital divide in Bloemfontein. 2. To analyse the impact of limited internet access on education and employment. 3. To evaluate the scalability and cost-effectiveness of streetlight-based Wi-Fi. 4. To explore how improved internet access can promote digital inclusion and socioeconomic development.
2.4	Research methodology
2.4.1	Research participants and their age brackets (where applicable, e.g. 10 Students from Civil Engineering Department)
	200 community members (age 18-50) and 100 Motheo tvet students (age 18 – 30)
2.4.2	How will participants be selected/sampled?
	purposive sampling method targeting Residents in underserved urban communities, students, Local business owners and informal traders.
2.4.3	Research site(s) (e.g. Borong Construction Site) Please list
	Rocklands, Bochabela and Motheo TVET College – Bloemfontein Campus
2.4.4	Data collection instruments (e.g. questionnaire(s)/interview schedule(s)/observation schedule(s)/artefacts/other)
	Observation Schedule This observation will observe infrastructure, power availability, streetlight positions, and public space usability. Questionnaire To gather quantitative and qualitative data on internet accessibility, digital needs, and opinions about public Wi-Fi via smart streetlights.

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2.4.5	Data collection procedure (Please outline WHEN, WHERE, and HOW data will be collected)
	The data will take over 3 weeks period The data will be collected at Rockland, Bochabela, and Motheo tvet college The data will be collected with observation schedules and questionnaires

3. PROPOSED PLAN OF STUDY/RESEARCH

	Set out your intended plan of work for the research, indicating important target dates necessary to meet your proposed deadlines	
	ACTIVITY	DATE

4. ETHICAL ISSUES AND RISK ASSESSMENT

In order to assess whether your proposed research is ethically compliant, ethics risks are categorised into four categories:

(1) Research involving minor risk

The likelihood of projected harm or inconvenience in the research is not greater than that experienced in daily life.

(2) Research involving low risk

Research in which the only anticipatable risk is one of potential awkwardness or discomfort to the participants.

(3) Research involving medium risk

Research in which there is a possible risk of harm or discomfort, but where appropriate steps can be taken to lessen or moderate overall risk.

(4) Research involving high risk

Research in which there is a real and foreseeable risk of harm and discomfort, which may lead to a serious adverse event if not managed in a responsible manner.

4.1	Will human research participants be used in your study? <i>Please mark with an X or V in the Yes/No/N/A box</i>	Yes	No	N/A
4.2	If yes, does the research study involve any of the following:			
	a) Children or youth under the age of 18 (Attach parental consent letter)		no	
	b) Individuals living with disabilities (physical, mental and/or sensory) (Attach consent letter of legal guardian)	yes		
	c) Individuals that might find it difficult to make independent and informed decisions for socio, economic, cultural, political and/or medical reasons		no	

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d) Communities that might be considered vulnerable, thus finding it difficult to make independent and informed decisions for socio, economic, cultural, political and/or medical reasons	yes		
e) Individuals who might be vulnerable for age related reasons e.g. the elderly	yes		
f) Individuals whose spoken language differs from the language used for the research (Make sure you translate your consent form and participant information sheet in the participants' first language – you should also have an interpreter if you do interviews – describe it below the table)	yes		
g) Women considered to be vulnerable (pregnancy, victimisation, marginalised etc.)	yes		
h) Other (Please explain):			

4.3	Will data collection involve any of the following:	Yes	No	N/A
a)	Access to confidential data without prior permission of participants		no	
b)	Participants expected to commit an act which might reduce self-respect or cause them to experience shame, embarrassment, or regret		no	
c)	Expose participants to worrying or upsetting questions or to processes which may have disagreeable or harmful side effects		no	
d)	The use of stimuli, errands or procedures which may be experienced as stressful, harmful, or hostile		no	
e)	Any use of materials risky to human beings		no	
4.4 If you answered "Yes", to any of the previously mentioned, explain (attach as an appendix) and justify. Explain, too, what steps you will take to minimise the potential stress/harm. (Please indicate if it is not applicable to your study)				

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4.5 Confidentiality of participants' identity				
4.5.1	Will the identity and privacy of participants be protected through pseudonyms or other forms of identification and the use of an informed consent form, which specifies (in a language that participants will understand): <i>Place an 'X' or '✓' in the Yes/No box</i>	YES	NO	N/A
4.5.2	Please note that participants should be informed about the following (where applicable)			
a)	The purpose/s of the research and how it is conducted	yes		
b)	The researcher, project leader and supervisor's identity, their institutional association and their contact details		no	
c)	Voluntary participation of participants	yes		
d)	Making sure that participants' responses will be treated in a confidential manner	yes		
e)	Be transparent about any possible limits on confidentiality which may apply	yes		
f)	Ensuring participants that they are free to withdraw from the research at any time without any negative or undesirable consequences to themselves	yes		
g)				
h)	How the findings of the study will have any benefits, or may receive as a result of their participation in the research	yes		
4.5.2 Please attach the proposed consent and assent documents prepared to address all the above, if not a full explanation is needed explaining how will participants be respected and protected.				

5. DOCUMENTS TO BE ATTACHED TO THE APPLICATION

The following documents must be attached as a prerequisite for approval to undertake research in the Department (where applicable)	
5.1	LS 262a approved by the FRIC (FEBIT)
5.2	Proof of registration/Funding received and funder reference details
5.3	Data collection instruments as identified under 2.4.4

6. DECLARATION BY THE APPLICANT

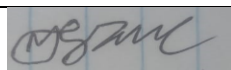
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I undertake to use the information that I acquire through my research, in a balanced and a responsible manner. I furthermore take note of, and agree to adhere to the following conditions (where applicable):

- a) I will schedule my research activities in consultation with the relevant Company or Organisation and research participants (where relevant);
- b) I agree that involvement of participants in my research is voluntary, and that participants have a right to decline to participate;
- c) I will obtain signed consent forms from participants prior to any engagement with them;
- d) I will inform participants about the use of recording devices such as tape-recorders and cameras, and participants will be free to reject them if they wish;
- e) I will honour the right of participants to privacy, anonymity, confidentiality and respect for human dignity at all times. Participants will not be identifiable in any way from the results of my research, unless written consent is obtained otherwise;
- f) All interviews (recordings) will be transcribed verbatim and analysed as per conventional data analysis techniques (example(s) of interview transcript to be included in final dissertation)
- g) I will adhere to the principles of rigorous data collection, analysis and interpretation consistent with the design of the study;
- h) I will keep a data trail for possible auditing purposes as well as the safe keeping of raw data for a period of three years after publication of the results;
- i) I will send the draft research findings to research participants before finalisation, in order to validate the accuracy of the information in the report;
- j) I will not use the resources of the university when I am conducting my research (such as stationery, photocopies, faxes, and telephones) and
- k) I will include a disclaimer in any report, publication or presentation arising from my research, that the findings and recommendations of the study do not represent the views of the Central University of Technology.
- l) Aside from laboratory as well as consumables or materials supplied by the university needed to complete practical projects which might be central to my study (dependent on study field), I will not use the resources of the University when I am conducting my research (such as stationery, photocopies, faxes, and telephones).
- m) All practical artefacts produced in support of my study using the university's laboratories, consumables, and materials will remain the property of the University.
- n) If I supplied my own materials and consumables, I will permit access to all practical projects or artefacts to the University for a period of three (3) years for exhibition purposes.
- o) All data collected for the research (including, but not limited to, completed questionnaires; statistical analysis performed on the data; interview audio-files/transcripts; artefacts/audio-visual materials; documents) will be kept safe at a designated space at the university for a period of at least three years. Computer files will be backed-up and password-protected.

I declare that all statements made in this application are true and accurate. I accept the conditions associated with the granting of approval to conduct research and undertake to abide by them.

**STUDENT SIGNATURE / PROJECT
LEADER SIGNATURE / SIGNATURE
OF RESEARCHER**



DATE

13 April 2025

7. DECLARATION BY SUPERVISOR(S) (where applicable)

I/We declare that I/we shall oversee the student's adherence to all statements as set out above.

SIGNATURE (Main supervisor)

SIGNATURE (Co-supervisor)

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DATE	
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FOR OFFICIAL USE**APPROVAL OF FEBIT ETHICAL COMMITTEE (FRIC)**

<i>Please tick relevant decision and provide conditions/reasons where applicable</i>		
Decision		<i>Please tick relevant option</i>
1.	Application approved	
2.	Ethical clearance number FRIC:	
3.	Application approved subject to certain conditions. <i>Specify conditions below</i>	
4.	Application not approved. <i>Provide reasons for non-approval below</i>	
SIGNATURE: Chairperson: Ethics committee		
DATE		

Cc Dean: FEBIT